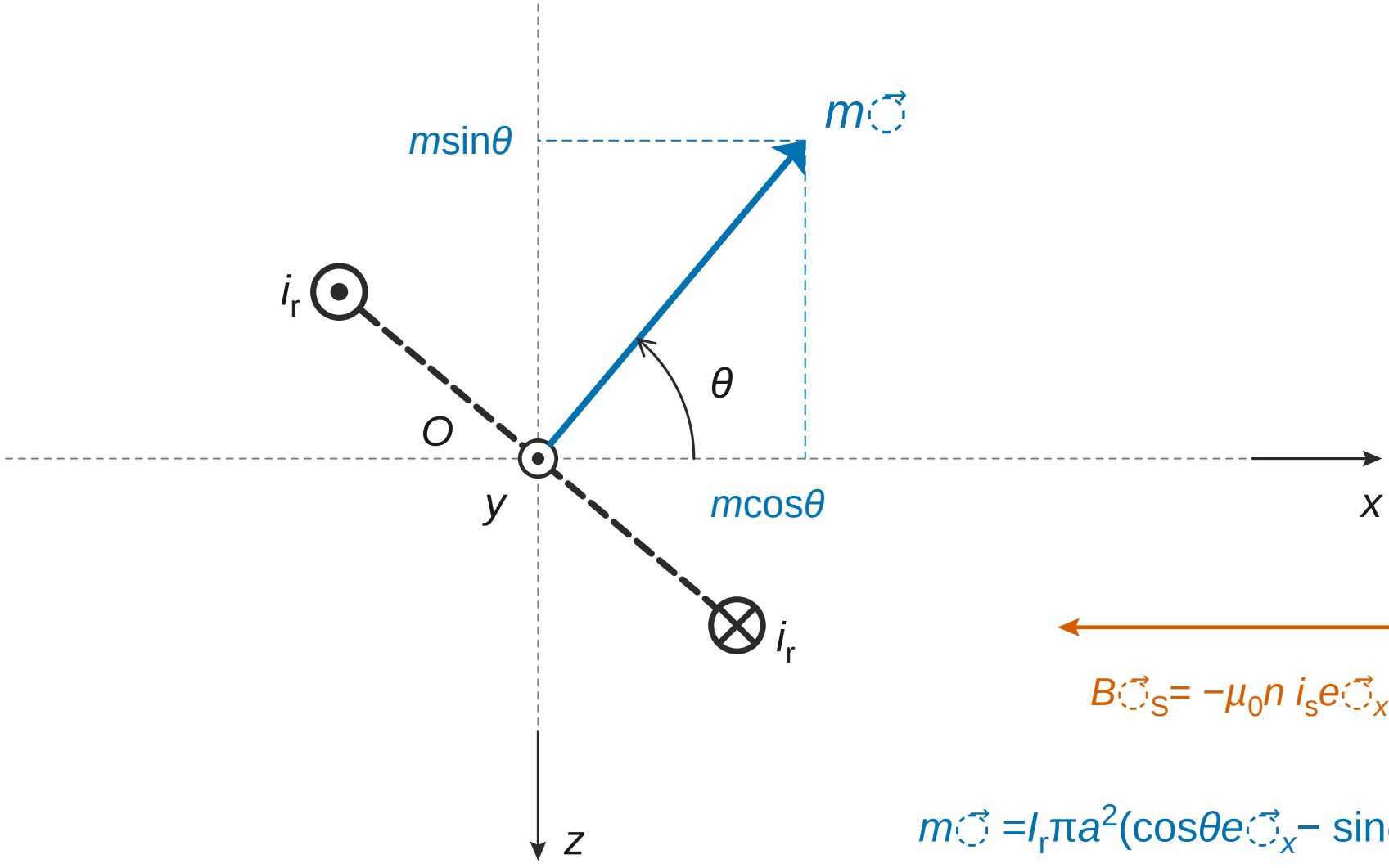

$$\vec{B}_S = -\mu_0 n\, i_s \vec{e}_x$$

$$\vec{m} = I_r \pi a^2 (\cos\theta \vec{e}_x - \sin\theta \vec{e}_z)$$